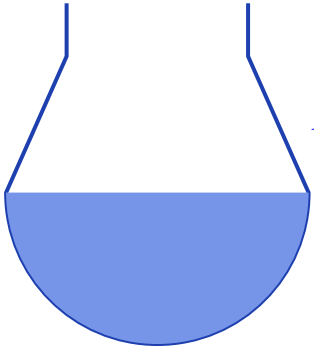
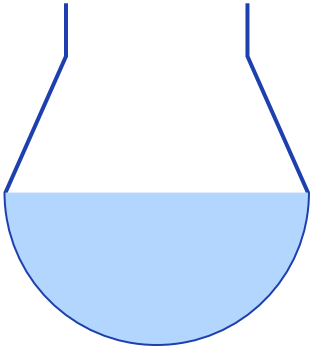


Add Pure Solvent

$+$

$\Delta V_{\text{solvent}}$

$=$



Mother Solution (Concentrated)

Initial Volume: V_1

Initial Molarity: C_1

Solute

Daughter Solution (Diluted)

Final Volume: $V_2 = V_1 + \Delta V > V_1$

Final Molarity: $C_2 < C_1$

C_2V_2

Conservation of Solute: $n_{\textbf{initial}} = n_{\textbf{final}} \iff \mathbf{C_1V_1 = C_2V_2}$